

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Semester V Examination, 2013-14

**Paper: CSB-503 — Discrete Mathematical Structure**

Time: Three hours

Maximum Marks: 70

*Instructions: Attempt any FIVE questions, The figures in right margin indicate the marks.***Question 1.**

- (a) Define successor of a set. Give two sets  $A$  and  $B$  such that  $A \in B$  and  $A \subset B$ . [4]
- (b) Define difference and symmetric difference of sets. Describe in general how these two set operations are achieved when sets are represented using bit strings. [5]
- (c) Define partition and covering of a set. Generate universal set  $E$  using  $A_1 = \{3, 4, 5\}$ ,  $A_2 = \{1, 2\}$ , and then give the partition of generated universal set using sets  $A_1, A_2$ . [5]

**Question 2.**

- (a) Consider the following relation  $R$  on set  $A = \{1, 2, 3, 4\}$ :

$$R = \{(a, b) \mid a \text{ divides } b\}$$

- (i) List all ordered pairs of relation  $R$ . [1]
- (ii) Represent relation  $R$  using digraph. [1]
- (iii) Give relation matrix of the relation  $R$ . [1]
- (iv) Find the matrix of  $R^{-1}$  from matrix of relation  $R$ . [1]
- (b) (i) Define partial order relation and poset. [2]
- (ii) Using only matrix operations show that the relation  $R$  defined in Question 2(a) is a partial order relation and hence  $(A, R)$  is a poset. [3]
- (iii) Draw the Hasse diagram of the above partial order relation. [1]
- (iv) Find the maximal and the minimal elements, if they exist, of the above poset  $(A, R)$ . [2]
- (v) Find the greatest and the least elements, if they exist, of the above poset  $(A, R)$ . [2]

**Question 3.**

- (a) Let  $A$  be a set having  $n$  elements. Show the following:
- (i) Total possible reflexive relations on  $A$  is  $2^{n(n-1)}$ . [2]
- (ii) Total possible symmetric relations on  $A$  is  $2^{n(n+1)/2}$ . [2]
- (iii) Total possible symmetric and reflexive relations on  $A$  is  $2^{n(n-1)/2}$ . [2]

- (iv) Total possible anti-symmetric relations on  $A$  is  $2^n \cdot 3^{n(n-1)/2}$ . [2]
- (b) Define number theoretic function, partial function, and primitive recursive function and give at least one example of each. [4]
- (c) Show that the total number of one-to-one functions from  $A$  to  $B$  is  $n(n-1)(n-2) \cdots (n-m+1)$ , where  $|A| = m$  and  $|B| = n$ . [2]

**Question 4.**

- (a) Prove the following without constructing truth table:
  - (i)  $\neg p \wedge (p \vee q) \rightarrow q$  [3]
  - (ii)  $[p \wedge (p \rightarrow q)] \rightarrow q$  [3]
- (b) Represent the following facts using propositions. Also find what rules of inference are used in each of these arguments: [6]
  - (i) Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
  - (ii) Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
  - (iii) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will get sunburned. Therefore, if I go swimming, then I will get sunburned.
- (c) What is the limitation of statement calculus and how is it removed in predicate calculus? [2]

**Question 5.**

- (a)
  - (i) Prove that in a group  $(G, \circ)$ , if every element is its own inverse, i.e.,  $a^2 = e$  for all  $a \in G$ , then  $G$  is an abelian group. [5]
  - (ii) Let  $(\mathbb{R}, +)$  denote the additive group of real numbers and  $(\mathbb{R}^*, \cdot)$  denote the multiplicative group of positive reals. Define a function  $f : \mathbb{R} \rightarrow \mathbb{R}^*$  as  $f(x) = a^x$ ,  $a > 0$ . Show that  $f$  is a homomorphism. [5]
- (b) Let  $f$  be a homomorphism from  $(\mathbb{Z}, +)$  to  $(\mathbb{Q}^*, \cdot)$ ,  $\mathbb{Q}^* = \mathbb{Q} - \{0\}$ . Suppose  $f(2) = 4$ . Find out  $f(6)$  and  $f(-8)$ . [4]

**Question 6.**

- (a) Model the round-robin tournament using a graph. [3]
- (b) What do the in-degree and out-degree of a vertex in the graph represent? [3]
- (c) Draw the graph of the following: [4]
  - (i)  $K_5$
  - (ii)  $C_5$
  - (iii)  $W_5$

- (iv)  $Q_3$
- (d) For which values of  $n$  are these graphs bipartite? Also find their chromatic number: [4]
  - (i)  $K_n$
  - (ii)  $C_n$
  - (iii)  $W_n$
  - (iv)  $Q_n$

**Question 7.**

- (a) Can five houses be connected to two utilities without connections crossing? [2]
- (b) A connected planar graph has 30 edges. If a planar representation of this graph divides the plane into 20 regions, how many vertices does this graph have? [4]
- (c) Find the complements of every element, if they exist, of  $(S_n, D)$  for  $n = 75$ , where  $S_n$  is the set of divisors of  $n$ , and  $D$  is the division relation. [5]
- (d) Determine whether the following statements are true, false, or not valid: [3]
  - (i)  $2 + 2 = 4$  iff  $1 + 1 = 2$ .
  - (ii) If  $1 + 1 = 3$ , then unicorns exist.
  - (iii)  $x + y = 2$ .